

Finding out Determinants (3 Methods)

- ① pivots
- ② Big Formula
- ③ Co-factor

1 Pivot Formula

Uses the method of elimination.

We know, $A = L \cdot U$

$$\Rightarrow \det(A) = \det(L) \cdot \det(U)$$

$$= 1 \cdot (\text{product of diagonal elements of } U)$$

$$= d_1 \cdot d_2 \cdot d_3 \cdot \dots \cdot d_n$$

A = any matrix

L = lower triangular matrix
(1 along principal diagonal)

U = Upper triangular matrix

If there are any row exchanges in the operation,

$$\det(A) = (\pm) d_1 \cdot d_2 \cdot d_3 \cdot \dots \cdot d_n$$

$$88 - = \left(\frac{01}{E} \times L \times L \times L \right) - = (A) \text{ to } 0$$

Example:

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 1 & 2 & 4 & 2 \\ 2 & 7 & 5 & 2 \\ 1 & 4 & -6 & 3 \end{bmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 0 & 2 & -8 & 2 \end{vmatrix} \quad \begin{aligned} r_4 &= r_4 - r_1 \\ r_3 &= r_3 - 2r_1 \\ r_2 &= r_2 - r_1 \end{aligned}$$

$$= \begin{vmatrix} 1 & 2 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 2 & -8 & 2 \end{vmatrix} \quad \text{row exchange } r_2 \text{ and } r_3$$

$$= \begin{vmatrix} 1 & 2 & 2 & 1 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -\frac{26}{3} & 2 \end{vmatrix} \quad r_4 = r_4 - (r_2 \times \frac{2}{3})$$

$$= - \begin{vmatrix} \textcircled{1} & 2 & 2 & 1 \\ 0 & \textcircled{3} & 1 & 0 \\ 0 & 0 & \textcircled{2} & 1 \\ 0 & 0 & 0 & \textcircled{\frac{19}{3}} \end{vmatrix} \quad r_4 = r_4 - (r_3 \times \frac{26}{6})$$

$$\therefore \det(A) = - (1 \times 3 \times 2 \times \frac{19}{3}) = -38$$

Example:

$$B = \begin{bmatrix} x & y & y & y & y \\ y & x & y & y & y \\ y & y & x & y & y \\ y & y & y & x & y \\ y & y & y & y & x \end{bmatrix} \xrightarrow{R_1 = R_1 + R_2 + R_3 + R_4 + R_5} \begin{bmatrix} x+4y & x+4y & x+4y & x+4y & x+4y \\ y & x & y & y & y \\ y & y & x & y & y \\ y & y & y & x & y \\ y & y & y & y & x \end{bmatrix}$$

$$(x+4y) \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ y & x & y & y & y \\ y & y & x & y & y \\ y & y & y & x & y \\ y & y & y & y & x \end{bmatrix} \xrightarrow{r_i = r_i - y \cdot r_1} (x+4y) \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & x-y & 0 & 0 & 0 \\ 0 & 0 & x-y & 0 & 0 \\ 0 & 0 & 0 & x-y & 0 \\ 0 & 0 & 0 & 0 & x-y \end{bmatrix}$$

$$\therefore \det(B) = (x+4y)(x-y)^4$$

2. The Big Formula

In the machines, usually the softwares uses elimination method to find determinants.

In comparison with the pivots, we change the entries of the matrices.

So, the big formula derives a single explicit formula directly from the entries of a matrix.

Thus, this formula has $n!$ terms.

So, if we have a matrix,

$$\begin{aligned} \begin{vmatrix} a & b \\ c & d \end{vmatrix} &= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} \quad (\text{row exchange}) \\ &= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} - \begin{vmatrix} c & 0 \\ 0 & b \end{vmatrix} = ad - bc \\ &\quad \underbrace{\hspace{10em}}_{2! \text{ terms}} \end{aligned}$$

Without changing the values of a matrix

$n \times n$ matrix $\rightarrow$ $n!$ terms

3 Determinants by co-factors

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Approach row-wise (only 1)

$$\begin{vmatrix} \boxed{a_{11}} & a_{12} & a_{13} \\ a_{21} & \boxed{a_{22}} & \boxed{a_{23}} \\ a_{31} & \boxed{a_{32}} & \boxed{a_{33}} \end{vmatrix}$$

co-factor of

a_{11}

(C_{11})

$$+ \begin{vmatrix} -a_{11} & \boxed{a_{12}} & a_{13} \\ a_{21} & \boxed{a_{22}} & \boxed{a_{23}} \\ a_{31} & \boxed{a_{32}} & \boxed{a_{33}} \end{vmatrix}$$

co-factor of

a_{12}

(C_{12})

$$+ \begin{vmatrix} a_{11} & a_{12} & \boxed{a_{13}} \\ a_{21} & \boxed{a_{22}} & \boxed{a_{23}} \\ a_{31} & \boxed{a_{32}} & \boxed{a_{33}} \end{vmatrix}$$

co-factor of

a_{13}

(C_{13})

$$\det(A) = a_{11}C_{11} - a_{12}C_{12} + a_{13}C_{13}$$

sign, a_{ij}

$i \rightarrow \text{row}$
 $j \rightarrow \text{column}$

if $i+j = \text{even}$, sign +ve
 $i+j = \text{odd}$, sign -ve

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

Co-factor along any row, $C_{ij} = (-1)^{i+j} (\det C_{ij})$

$$\det(A) = a_{11}(a_{22} \cdot a_{33} - a_{23} \cdot a_{32}) - a_{12}(a_{21} \cdot a_{33} - a_{23} \cdot a_{31}) + a_{13}(a_{21} \cdot a_{32} - a_{22} \cdot a_{31})$$

for a $n \times n$ matrix,

$$\det(A) = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} + \dots + a_{1n}c_{1n}$$

Example:

$$A = \begin{bmatrix} x & y & 0 & 0 & 0 \\ 0 & x & y & 0 & 0 \\ 0 & 0 & x & y & 0 \\ 0 & 0 & 0 & x & y \\ y & 0 & 0 & 0 & x \end{bmatrix}$$

$$A^T = \begin{bmatrix} x & 0 & 0 & 0 & y \\ y & x & 0 & 0 & 0 \\ 0 & y & x & 0 & 0 \\ 0 & 0 & 0 & x & y \\ 0 & 0 & y & x & 0 \end{bmatrix}$$

$$A^T = x \begin{vmatrix} x & 0 & 0 & 0 \\ y & x & 0 & 0 \\ 0 & y & x & 0 \\ 0 & 0 & y & x \end{vmatrix} + y \begin{vmatrix} y & x & 0 & 0 \\ 0 & y & x & 0 \\ 0 & 0 & x & y \\ 0 & 0 & 0 & y \end{vmatrix}$$

$$= x \cdot x^4 + y \cdot y^4$$

$$= x^5 + y^5$$

False Expansion Theorem

If A is an $n \times n$ matrix, then

$$a_{i1} \cdot C_{k1} + \dots + a_{in} \cdot C_{kn} = 0 \quad [i \neq k]$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

If we multiply the co-factors of the elements of a certain row with other row elements, its sum will be 0.

$$a_{21} \cdot C_{11} + a_{22} \cdot C_{12} + a_{23} \cdot C_{13} = 0$$

$$a_{31} \cdot C_{11} + a_{32} \cdot C_{12} + a_{33} \cdot C_{13} = 0$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Proof:

From the previous part,

$$a_{11} \cdot C_{11} + \dots + a_{1n} \cdot C_{1n} = \det(A)$$

$$a_{21} \cdot C_{21} + \dots + a_{2n} \cdot C_{2n} = \det(A)$$

$$a_{31} \cdot C_{31} + \dots + a_{3n} \cdot C_{3n} = \det(A)$$

$$\left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] \left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] \left[\text{When } i = k \right]$$

Let B be a matrix obtained from A by replacing the k -th row of A with the i -th row of A .

So, two rows of B are same,

$$\therefore \det(B) = 0.$$

$$A = \begin{bmatrix} \boxed{a \quad b} \\ \boxed{c \quad d} \end{bmatrix} \begin{array}{l} \rightarrow i\text{-th row} \\ \rightarrow k\text{-th row} \end{array}$$

$$B = \begin{bmatrix} a & b \\ a & b \end{bmatrix} \quad \det(B) = 0$$

Now, if we were to compute $\det(B)$ by expanding along k -th row, (co-factors of B matrix*)

$$b_{k1} \cdot C_{k1}^* + b_{k2} \cdot C_{k2}^* + \dots + b_{kn} \cdot C_{kn}^* = 0 \quad \text{--- (i)}$$

$$k\text{-th row of } B = i\text{-th row of } A$$

$$\therefore a_{i1} \cdot C_{k1}^* + a_{i2} \cdot C_{k2}^* + \dots + a_{in} \cdot C_{kn}^* = 0 \quad \text{--- (ii)}$$

The two matrices A and B differ only in the k -th row, this doesn't change the co-factors of the i -th row

C_{ij}^* is the co-factor of b_{ij} in B

As we are picking k -th row of B and i -th row of A , their co-factors will be same.

$$(A) \cdot (B) = 12 \cdot 10 = 120$$

$$0 = 120 - 120 = 0$$

→ Co-factor of k th and i th rows are same
 [a from (ii), c from (i)]

$$a_{i1} \cdot c_{k1} + a_{i2} \cdot c_{k2} + \dots + a_{in} \cdot c_{kn} = 0$$

$$A = \begin{bmatrix} a & b \\ \textcircled{c} & \textcircled{d} \\ -b & a \end{bmatrix} \quad B = \begin{bmatrix} a & b \\ \textcircled{a} & \textcircled{b} \\ -b & a \end{bmatrix} \rightarrow k\text{-th row}$$

(ii) → Co-factors

$$a \cdot b - b \cdot a = 0$$

[proved]

It can be also proved using $\begin{bmatrix} c & d \\ d & c \end{bmatrix}$

$$\therefore \det(A) = \vec{a}_i \cdot \vec{c}_i$$

$$0 = \vec{a}_i \cdot \vec{c}_k \quad [i \neq k]$$

$$(A \cdot (\text{cof}(A))^T)_{ii} = a_i \cdot c_i = \det(A)$$

$$(A \cdot (\text{cof}(A))^T)_{ik} = 0$$

Example:

$$I(A^{-1}) = {}^T(A^{-1}) \cdot A$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{cof}^T(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A \cdot (\text{cof}^T(A))^T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

$$= \begin{bmatrix} \det(A) & 0 \\ 0 & \det(A) \end{bmatrix}$$

$$= \det(A) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \det(A) \cdot I$$

$$A \cdot (\text{cof}^T A)^T = (\det A) I$$

$$A \cdot (\text{cof } A)^T = (\det A) I$$

$$A \cdot \frac{(\text{cof } A)^T}{\det(A)} = I = A \cdot A^{-1}$$

$$\Rightarrow A^{-1} = \frac{(\text{cof } A)^T}{\det(A)}$$

$$A^{-1} = \frac{1}{\det(A)} (\text{cof } A)^T$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (A)^T b =$$

$$I (A)^T b =$$

$$I (A)^T b = (A)^T b$$

Example:

$$B = \begin{bmatrix} 7 & 1 & 2 \\ 4 & -2 & -5 \\ 9 & 8 & -3 \end{bmatrix}$$

$$\det(B) = 439$$

$$\text{cof}(B) = \begin{bmatrix} 46 & -33 & 50 \\ 27 & -48 & -47 \\ 1 & 47 & -18 \end{bmatrix}$$

$$\therefore B^{-1} = \frac{1}{439} \cdot (\text{cof } B)^T$$

$$= \frac{1}{439} \begin{bmatrix} 46 & -33 & 50 \\ 27 & -48 & -47 \\ 1 & 47 & -18 \end{bmatrix}$$

Application of Determinant

① Cramer's Rule

$$A = \begin{bmatrix} 2 & 0 & 2 \\ 2 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$X = \begin{bmatrix} 15 \\ 25 \\ 35 \end{bmatrix}$$

$$\frac{\det B}{\det A} = 15$$

$$\frac{\det B}{\det A} = 25$$

$$\frac{\det B}{\det A} = 35$$

$$\frac{\det B}{\det A} = 35$$

Application of Determinant

① Cramer's Rule

$$A = \begin{bmatrix} 2 & 6 & 2 \\ 1 & 4 & 2 \\ 5 & 9 & 0 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$Ax = b$, what x creates this result?

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\det(A) = 2$$

$$x_1 = \frac{\det B_1}{\det A}$$

$$x_2 = \frac{\det B_2}{\det A}$$

$$x_3 = \frac{\det B_3}{\det A}$$

$$x_n = \frac{\det B_n}{\det A}$$

$$B_i = \begin{bmatrix} \uparrow & (n-1) \text{ other columns} \\ b & \text{of } A \\ \downarrow \end{bmatrix}.$$

$i \rightarrow$ column no. of b ,

for B_2 , b will be the 2nd column of A

$$\therefore B_1 = \begin{vmatrix} 0 & 6 & 2 \\ 0 & 4 & 2 \\ 1 & 9 & 0 \end{vmatrix} \quad \det(B_1) = 4$$

$$B_2 = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 0 & 2 \\ 5 & 1 & 0 \end{vmatrix} \quad \det(B_2) = -2.$$

$$B_3 = \begin{vmatrix} 2 & 6 & 0 \\ 1 & 4 & 0 \\ 5 & 9 & 1 \end{vmatrix} \quad \det(B_3) = 2$$

$$\lambda_1 = \frac{4}{2} = 2$$

$$\lambda_2 = \frac{-2}{2} = -1 \quad \therefore \lambda = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\lambda_3 = \frac{2}{2} = 1$$